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MAT251_TW | Chapter 7: Sequences and Series

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Question **1**

Complete

Not graded

When determining the convergence of

$$\sum_{n=1}^{\infty} \frac{5n^2 + 2n - 1}{3(n^2 + 2)^2}$$

by the limit comparison test, which of the following is the best p-series to use?

Select one:

a. $\sum_{n=1}^{\infty} \frac{1}{n^4}$

b. $\sum_{n=1}^{\infty} \frac{1}{n^2}$

c. $\sum_{n=1}^{\infty} \frac{1}{n^{1/2}}$

d. $\sum_{n=1}^{\infty} \frac{1}{n^0}$

$$\frac{5n^2 + 2n - 1}{3\left(n^2 + 2\right)^2} = \frac{5n^2 + 2n - 1}{3n^4 + 12n^2 + 12}$$

The numerator and denominator each behave like their highest powers.

Therefore, $\frac{5n^2 + 2n - 1}{3\left(n^2 + 2\right)^2}$ behaves like $\frac{n^2}{n^4} = \frac{1}{n^2}$.

The correct answer is: $\sum_{n=1}^{\infty} \frac{1}{n^2}$

Question **2**

Complete

Not graded

Using the limit comparison test, determine whether $\sum_{n=1}^{\infty} \frac{5}{n^2+3}$ diverges or converges.

Select one:

- a. The series diverges.
- b. The series converges.
- c. The test is inconclusive.

Using $b_n = 1/n^2$, you get

$$L = \lim_{n\to\infty} \frac{a_n}{b_n}$$

$$= \lim_{n\to\infty} \frac{\frac{5}{n^2+3}}{\frac{1}{n^2}}$$

$$= \lim_{n\to\infty} \frac{5n^2}{n^2+3}$$

$$= \lim_{n\to\infty} \frac{5n^2}{n^2}$$

$$= 5.$$

Because $0<L<\infty$, you can conclude that $\sum_{n=1}^{\infty} \frac{5}{n^2+3}$ converges by the limit comparison test with $\sum_{n=1}^{\infty} \frac{1}{n^2}$ (it is a p-series where $p > 1$).

The correct answer is: The series converges.

Question **3**

Complete

Not graded

Which of the following is the best choice for b_n when using the limit comparison test on the series $\sum_{n=1}^{\infty} \frac{3}{n^2+2}$?

Select one:

- a. $\frac{1}{n^3}$
- b. n^2+2
- c. $\frac{3}{2}$
- d. $\frac{1}{n^2}$

This is a p-series where $p = 2$, as desired.

In addition, all terms are positive. Finally,

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{\frac{3}{n^2+2}}{\frac{1}{n^2}}$$

$$= \lim_{n\to\infty} \frac{3n^2}{n^2+2}$$

$$= \lim_{n\to\infty} \frac{6n}{2n}$$

$$= 3,$$

which is positive and finite. Therefore,

because $\sum_{n=1}^{\infty} \frac{1}{n^2}$

is known to converge (it is a p-series

where $p > 1$), $\sum_{n=1}^{\infty} \frac{3}{n^2+2}$ also converges

by the limit comparison test with $\sum_{n=1}^{\infty} \frac{1}{n^2}$.

The correct answer is: $\frac{1}{n^2}$

Question **4**

Complete

Not graded

Given $\sum_{n=1}^{\infty} a_n$, where $a_n = \frac{\sqrt{x}}{x - \ln x}$, and $\sum_{n=1}^{\infty} b_n$, where $b_n = \frac{1}{\sqrt{x}}$, which of the following ratios will have fewer terms in the denominator than in the numerator?

Select one:

a. $\lim_{n\to\infty} \frac{a_n}{b_n}$

b. $\lim_{n\to\infty} \frac{b_n}{a_n}$

This ratio produces a monomial in the denominator and two terms in the numerator.

$$\lim_{n\to\infty} \frac{b_n}{a_n} = \left(\frac{1}{\sqrt{x}} \cdot \frac{x - \ln x}{\sqrt{x}}\right)$$
$$= \lim_{n\to\infty} \frac{x - \ln x}{x}$$
$$= \lim_{n\to\infty} \left(1 - \frac{\ln x}{x}\right)$$
$$= 0$$

The correct answer is: $\lim_{n\to\infty} \frac{b_n}{a_n}$

Question **5**

Complete

Not graded

Given $\sum_{n=1}^{\infty} a_n$, where $a_n = \frac{3n^3}{5n^5 - 2\sqrt{n}}$, and $\sum_{n=1}^{\infty} b_n$, where $b_n = \frac{1}{n^2}$, which of the following ratios will have fewer terms in the denominator than in the numerator?

Select one:

- a. $\lim_{n\to\infty} \frac{a_n}{b_n}$
- b. $\lim_{n\to\infty} \frac{b_n}{a_n}$

This ratio produces a monomial in the denominator and two terms in the numerator.

This limit is much easier to evaluate.

$$\lim_{n\to\infty} \frac{b_n}{a_n} = \lim_{n\to\infty} \frac{\frac{1}{n^2}}{\frac{3n^3}{5n^5 - 2\sqrt{n}}}$$

$$= \lim_{n\to\infty} \frac{5n^5 - 2\sqrt{n}}{3n^5}$$

$$= \lim_{n\to\infty} \left(\frac{5}{3} - \frac{2}{3n^{9/2}} \right)$$

$$= \frac{5}{3}$$

The correct answer is: $\lim_{n\to\infty} \frac{b_n}{a_n}$

Question **6**

Complete

Not graded

Determine the behavior of

$\sum_{n=1}^{\infty} \frac{n}{2+e^n}$, by using $\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{n}{e^n}$ in the limit comparison test.

Select one:

- a. The series converges.
- b. The series diverges.
- c. The test is inconclusive.

$$\lim_{n\to\infty} \frac{b_n}{a_n} = \lim_{n\to\infty} \left(\frac{n}{e^n} \cdot \frac{2+e^n}{n} \right)$$

$$= \lim_{n\to\infty} \frac{2+e^n}{e^n}$$

$$= \lim_{n\to\infty} \left(\frac{2}{e^n} + \frac{e^n}{e^n} \right)$$

$$= 0+1$$

$$= 1.$$

Therefore, $\sum_{n=1}^{\infty} \frac{n}{2+e^n}$ converges by the limit comparison test with $\sum_{n=1}^{\infty} \frac{n}{e^n}$.

The correct answer is: The series converges.

Question **7**

Complete

Not graded

$$\sum_{n=1}^{\infty} \frac{5}{2n^2 - 3}$$

Using the limit comparison test, determine whether the series converges or diverges.

Select one:

- a. The series converges.
- b. The series diverges.
- c. The test is inconclusive.

Because only the highest powers of n in the numerator and denominator affect the "spirit" of the series, it has the same spirit as $\sum_{n=1}^{\infty} \frac{1}{n^2}$.

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{\frac{5}{2n^2 - 3}}{\frac{1}{n^2}}$$

$$= \lim_{n\to\infty} \frac{5n^2}{2n^2 - 3}$$

$$= \lim_{n\to\infty} \frac{10n}{4n}$$

$$= \frac{10}{4}$$

$$= \frac{5}{2}$$

Because this limit is positive and finite, $\sum_{n=1}^{\infty} \frac{5}{2n^2 - 3}$ converges by the limit comparison test with $\sum_{n=1}^{\infty} \frac{1}{n^2}$.

The correct answer is: The series converges.

Question **8**

Complete

Not graded

Determine the behavior of

$\sum_{n=1}^{\infty} \frac{\sqrt{n} + ln\, n^2}{2n}$, using the limit comparison test.

Select one:

- a. The series converges.
- b. The series diverges.
- c. The test is inconclusive.

Use $\lim_{n\to\infty}\frac{a_n}{b_n}$ since there are more terms in the numerator than the denominator.

Using $b_n = \frac{1}{\sqrt{n}}$ and $\lim_{n\to\infty}\frac{a_n}{b_n}$ produces

$$\lim_{n\to\infty}\frac{a_n}{b_n} = \lim_{n\to\infty}\frac{\frac{\sqrt{n}+\ln n^2}{2n}}{\frac{1}{\sqrt{n}}}$$

$$= \lim_{n\to\infty}\frac{n+\sqrt{n}(2\ln n)}{2n}$$

$$= \frac{1}{2} + \lim_{n\to\infty}\frac{\ln n}{\sqrt{n}}.$$

Using L'Hôpital's Rule on this last limit produces

$$\lim_{n\to\infty}\frac{a_n}{b_n} = \frac{1}{2} + \lim_{n\to\infty}\frac{\ln n}{\sqrt{n}}$$

$$= \frac{1}{2} + \lim_{n\to\infty}\frac{\frac{1}{n}}{\frac{1}{2\sqrt{n}}}$$

$$= \frac{1}{2} + \lim_{n\to\infty}\frac{2\sqrt{n}}{n}$$

$$= \frac{1}{2} + \lim_{n\to\infty}\frac{1}{\sqrt{n}}$$

$$= \frac{1}{2} + 2(0)$$

$$\frac{1}{2}.$$

Therefore, $\sum_{n=1}^{\infty}\frac{\sqrt{n}+\ln n^2}{2n}$ diverges by the limit comparison test with $\sum_{n=1}^{\infty}\frac{1}{\sqrt{n}}$.

The correct answer is: The series diverges.

Question **9**

Complete

Not graded

When determining the convergence of

$$\sum_{n=1}^{\infty} \frac{2\sqrt{n+1}}{n}$$

by the limit comparison test, which of the following is the best p-series to use?

Select one:

a. $\sum_{n=1}^{\infty} n^{1/2}$

b. $\sum_{n=1}^{\infty} \frac{1}{n^2}$

c. $\sum_{n=1}^{\infty} \frac{1}{n}$

d. $\sum_{n=1}^{\infty} \frac{1}{n^{1/2}}$

The numerator and denominator each behave like their highest powers.

Therefore, $\frac{2\sqrt{n+1}}{n}$ behaves like

$$\frac{\sqrt{n}}{n} = \frac{n^{1/2}}{n} = \frac{1}{n^{1/2}}.$$

The correct answer is: $\sum_{n=1}^{\infty} \frac{1}{n^{1/2}}$

Question **10**

Complete

Not graded

$$\sum_{n=1}^{\infty} \frac{\sqrt{n}}{2n^2 - 1}$$

Using the limit comparison test, determine whether the series converges or diverges.

Select one:

- a. The series converges.
- b. The series diverges.
- c. The test is inconclusive.

Because only the highest powers affect the behavior of a_n as n becomes infinite, a_n behaves like the quotient of the highest powers of the numerator and denominator

(i.e., $\frac{\sqrt{n}}{n^2} = \frac{n^{1/2}}{n^2} = \frac{1}{n^{3/2}}$).

$\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$ is a p-series where $p = \frac{3}{2} > 1$,

and is therefore known to converge.

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{\frac{1}{n^{3/2}}}{\frac{\sqrt{n}}{2n^2-1}}$$

$$= \lim_{n\to\infty} \frac{\frac{1}{n^{3/2}}}{\frac{n^{1/2}}{2n^2-1}}$$

$$= \lim_{n\to\infty} \frac{2n^2-1}{n^2}$$

$$= \lim_{n\to\infty} \left(2 - \frac{1}{n^2}\right)$$

$$= 2$$

Because this limit is positive and finite,

we conclude that $\sum_{n=1}^{\infty} \frac{\sqrt{n}}{2n^2-1}$ converges

by the limit comparison test with

$\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$.

The correct answer is: The series converges.

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